Experimental Expansion of Brain Tumor MRI Classification Using EfficientNetB0, Multi-Dataset Training, and Grad-CAM Explainability

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School of Computing, FAST-NUCES
Degree: BSAI
Course: Artificial Neural Networks
Assignment: Assignment 3 - Experimentation and Expansion
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Due Date: 26-04-2026
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Abstract

Brain tumor classification from MRI images is an important medical image analysis task because correct classification can support early diagnosis and treatment planning. In this assignment, the work from the previous reproduction stage was extended with new experiments. The reproduced baseline used ResNet50 transfer learning for four-class brain tumor MRI classification. The four target classes were glioma, meningioma, no-tumor, and pituitary tumor.

The main objective of this assignment was to move beyond reproduction and perform experimental expansion. Three experiments were conducted. First, a frozen ResNet50 model was trained as a basic transfer learning baseline. Second, a fine-tuned ResNet50 model was trained as the reproduced improved baseline. Third, EfficientNetB0 was used as the proposed model, and the training data was expanded by combining two independent brain tumor MRI datasets.

The models were evaluated using accuracy, macro-F1 score, class-wise precision/recall/F1-score, normalized confusion matrices, training curves, Grad-CAM visualizations, and McNemar's statistical significance test. The proposed EfficientNetB0 model achieved the highest numerical performance on the Dataset 1 test set, with 94.50% accuracy and 94.34% macro-F1 score. It also achieved 98.67% accuracy and 98.58% macro-F1 score on the held-out test split of Dataset 2. However, the improvement over the fine-tuned ResNet50 baseline was small and was not statistically significant according to McNemar's test. Therefore, the proposed method is best described as an efficient and explainable expansion of the reproduced baseline rather than a fully proven superior model.

Keywords: Brain tumor classification, MRI, ResNet50, EfficientNetB0, transfer learning, Grad-CAM, CNN, medical image classification.

1. Introduction

Brain tumors are serious medical conditions that require accurate diagnosis. Magnetic resonance imaging, also called MRI, is commonly used for brain analysis because it provides detailed information about soft tissues. MRI scans can help doctors identify tumor location and tumor type, but manual inspection takes time and depends on expert experience. Deep learning can support this task by automatically learning useful patterns from MRI images.

In this project, the task is to classify brain MRI images into four classes:

- glioma
- meningioma
- no-tumor
- pituitary tumor

This is a multi-class classification problem. It is more detailed than binary tumor detection because the model must not only identify whether a tumor exists but also classify the tumor category. This makes the task more challenging, especially because some tumor classes can look similar in two-dimensional MRI slices.

In Assignment 1 and Assignment 2, the main focus was on understanding and reproducing an existing brain tumor MRI classification approach. Assignment 3 focuses on experimentation and expansion. The reproduced work is used as a base, and then new experiments are added to test possible improvements.

The main extension in this work is the use of EfficientNetB0 with an additional MRI dataset. EfficientNetB0 was selected because it is a lightweight and efficient convolutional neural network. It can provide strong classification performance while using fewer trainable parameters than a larger fine-tuned model. A second dataset was added to increase the amount and variety of training data.

The main research questions in this report are:

1. Does fine-tuning ResNet50 improve performance compared to using it only as a frozen feature extractor?
2. Does EfficientNetB0 with expanded training data give better numerical results than the ResNet50 baselines?
3. Does the proposed method improve efficiency in terms of trainable parameters and training time?
4. Which tumor classes are still difficult for the model?
5. Are the observed improvements statistically significant?

The rest of the report is organized as follows. Section 2 gives the background and paper summary. Section 3 explains the reproduction summary. Section 4 presents the proposed method. Section 5 describes the experimental setup. Section 6 presents the results and analysis. Section 7 discusses the findings and limitations. Section 8 gives the conclusion. Section 9 lists the references.

2. Background and Paper Summary

2.1 Brain Tumor MRI Classification

Brain tumor MRI classification is a supervised medical image classification task. The input is an MRI image, and the output is a class label. In this project, each MRI image belongs to one of four classes: glioma, meningioma, no-tumor, and pituitary tumor.

Traditional machine learning approaches usually require handcrafted features such as texture, edge, shape, or intensity-based features. These methods can work, but they depend heavily on the selected features. If the handcrafted features do not capture important tumor patterns, the final classifier may perform poorly.

Convolutional Neural Networks reduce this problem by learning useful features directly from images. CNNs learn low-level features such as edges and textures in early layers, and more complex image patterns in deeper layers. For MRI classification, CNNs can learn tumor-related visual patterns such as local structure, shape, intensity changes, and texture differences.

2.2 Summary of the Reproduced Work

The reproduced work focused on supervised brain tumor MRI classification using transfer learning. The target was to classify MRI images into four categories. The previous baseline used a ResNet50-based CNN model pretrained on ImageNet.

The main idea behind the reproduced work was that a pretrained CNN can provide useful visual features, and these features can be adapted for MRI classification. In the first stage, the pretrained backbone can be frozen and only the final classifier can be trained. In the second stage, deeper layers can be fine-tuned so that the model learns features that are more specific to MRI images.

The reproduced approach showed that fine-tuning improved performance compared to a frozen transfer learning baseline. However, there were still some limitations:

- the main experiments were performed on one dataset,
- the fine-tuned ResNet50 model used many trainable parameters,
- glioma and meningioma were still confused in several cases,
- explainability was needed to understand which image regions influenced predictions.

2.3 ResNet50 Transfer Learning

ResNet50 is a deep convolutional neural network that uses residual connections. These skip connections help the model train deeper networks by improving gradient flow. ResNet50 is widely used as a transfer learning model because it has strong pretrained features from ImageNet.

In this project, ResNet50 was used in two ways. First, it was used as a frozen feature extractor. In this case, the convolutional base was not trained, and only the classification head was updated. Second, it was fine-tuned by unfreezing the last 50 layers. This allowed the model to adapt more strongly to the MRI dataset.

2.4 EfficientNetB0

EfficientNetB0 is a convolutional neural network designed to balance accuracy and efficiency. It uses compound scaling, where depth, width, and resolution are scaled in a balanced way. This makes EfficientNet models useful when good performance is needed with fewer parameters.

EfficientNetB0 was used as the proposed model in this assignment. The reason for choosing it was that it is lighter than ResNet50 but still strong for image classification. In this work, EfficientNetB0 was trained with combined data from two MRI datasets.

2.5 Grad-CAM Explainability

In medical image classification, it is useful to know why a model made a certain prediction. A model that gives only a label is less useful than a model that also provides some explanation of its decision.

Grad-CAM is a visualization method that highlights the regions of an image that contributed most to a model's prediction. It uses gradients from the final convolutional layer to produce a heatmap. In this project, Grad-CAM was used to qualitatively inspect whether the proposed model was focusing on meaningful MRI regions or irrelevant background areas.

3. Reproduction Summary

The reproduction stage was based on a ResNet50 transfer learning pipeline for brain tumor MRI classification. The same four classes were used: glioma, meningioma, no-tumor, and pituitary tumor. All images were resized to 224 × 224 pixels.

Before the final TensorFlow/Keras experiments, an earlier reproduction iteration was also performed using PyTorch. In that stage, a frozen ResNet50 model achieved 90.36% validation accuracy and 86.62% test accuracy, while a fine-tuned ResNet50 model reached 95.06% test accuracy. The main error pattern was confusion between glioma and meningioma.

For the final Assignment 3 pipeline, the reproduction and extension were implemented using TensorFlow/Keras with a stable tf.data input pipeline. Two main reproduction experiments were performed before the proposed extension.

3.1 Experiment 1: Frozen ResNet50 Baseline

In the first experiment, ResNet50 was used as a frozen feature extractor. The convolutional layers were not trained. Only the final classification head was trained for the four MRI classes. This experiment was used as the simplest transfer learning baseline. It tested whether general ImageNet features were enough for MRI classification.

3.2 Experiment 2: Fine-Tuned ResNet50 Baseline

In the second experiment, ResNet50 was fine-tuned. The last 50 layers were unfrozen and trained with a smaller learning rate. This allowed the model to adapt to MRI-specific patterns.

Fine-tuning improved performance compared to the frozen baseline. This showed that the MRI domain has features that are different from natural ImageNet images. Allowing the model to update deeper layers helped it learn better tumor-related features.

3.3 Main Findings from Reproduction

- Frozen ResNet50 gave a strong starting result.
- Fine-tuned ResNet50 performed better than frozen ResNet50.
- Fine-tuning helped the model adapt to MRI-specific features.
- No-tumor and pituitary classes were easier for the model.
- Glioma and meningioma were more commonly confused.

4. Proposed Method

The proposed method extends the reproduced baseline in two main ways. First, the model architecture was changed from ResNet50 to EfficientNetB0. This was done to test whether a more efficient CNN backbone could achieve similar or better performance with fewer trainable parameters. Second, an additional brain tumor MRI dataset was added. This was done to increase training data and give the model more variation during training.

4.1 Main Idea

The main idea of the proposed method is to use a lighter model and more training data to improve performance and efficiency. The reproduced baseline used ResNet50 on Dataset 1. The proposed method uses EfficientNetB0 and combines Dataset 1 with Dataset 2 for training.

4.2 Hypothesis

The main hypothesis is that EfficientNetB0 trained with combined MRI data from two datasets will achieve better numerical performance than the reproduced ResNet50 baselines. The secondary hypothesis is that EfficientNetB0 will be more efficient than fine-tuned ResNet50 because it uses fewer trainable parameters and requires less training time.

4.3 Proposed Pipeline

1. Load Dataset 1 and Dataset 2.
2. Verify folder structure and class names.
3. Standardize the class labels across both datasets.
4. Resize all images to 224 × 224.
5. Apply augmentation only to training images.
6. Train EfficientNetB0 using combined training data.

7. Evaluate the model on the Dataset 1 test set.
8. Evaluate the model on the held-out Dataset 2 test split.
9. Generate training curves, confusion matrices, and classification reports.
10. Use Grad-CAM for qualitative explainability.
11. Use McNemar's test for statistical comparison.

Table A: Experimental design summary

Experiment	Model	Training Data	Strategy	Purpose
Exp 1	ResNet50	Dataset 1	Frozen backbone	Basic transfer learning baseline
Exp 2	ResNet50	Dataset 1	Fine-tuned last 50 layers	Reproduced improved baseline
Exp 3	EfficientNetB0	Dataset 1 + Dataset 2	Fine-tuned last 30 layers	Proposed method

The proposed method changes both the model backbone and the training data. This is useful as an expansion experiment, but it is also a limitation. Since two things are changed at the same time, the improvement cannot be linked only to EfficientNetB0 or only to the additional dataset. A better ablation study in the future would train EfficientNetB0 on Dataset 1 only.

5. Experimental Setup

5.1 Environment

The experiments were implemented using TensorFlow/Keras. The notebook was executed in a Kaggle GPU environment. TensorFlow detected two Tesla T4 GPUs, but the final training pipeline was kept single-GPU compatible for stability.

Table B: Environment and global settings

Setting	Value
TensorFlow version	2.19.0
GPU environment	Kaggle Tesla T4 GPUs available
Batch size	32
Image size	224 × 224
Number of classes	4
Random seed	42

During early testing, multi-GPU training caused shape synchronization issues. The final pipeline used fixed image resizing and drop_remainder=True to keep batch shapes stable.

5.2 Dataset Description

Two brain tumor MRI datasets were used in this project. Dataset 1 was used as the main dataset for the reproduced baseline. Dataset 2 was added as the required additional dataset for Assignment 3.

Table C: Dataset summary

Dataset	Training Folder	Test Folder	Classes
Dataset 1: Brain Tumor MRI Dataset	5,600	1,600	glioma, meningioma, notumor, pituitary
Dataset 2: BT-MRI 4-Class Dataset	5,712	1,311	Glioma, Meningioma, No-tumor, Pituitary

The Dataset 1 training folder was split into 4,480 training images and 1,120 validation images. The Dataset 2 training folder was split into 4,570 training images and 1,142 validation images. For the proposed EfficientNetB0 experiment, the actual combined training set contained 9,050 images after the 80/20 split.

5.3 Class Labels and Label Standardization

Table D: Unified class labels

Class Index	Unified Class Name
0	glioma
1	meningioma
2	notumor
3	pituitary

Dataset 2 used labels such as "Glioma", "Meningioma", "No-tumor", and "Pituitary". These were checked against Dataset 1 labels. The alphabetical index order matched, so the labels were aligned before merging.

5.4 Preprocessing and Augmentation

All images were resized to 224 × 224 × 3. The preprocessing pipeline included image loading, resizing, conversion to float tensors, model-specific preprocessing, batching, shuffling for training data, and prefetching for faster training. Model-specific preprocessing was applied using Keras application preprocessing functions for ResNet50 and EfficientNetB0.

Data augmentation was applied only to the training split. The augmentation operations included random horizontal flip, random brightness adjustment, and random contrast adjustment. No augmentation was applied to validation or test data.

5.5 Model Settings and Training Configuration

Table E: Model settings

Model	Backbone	Fine-tuning Strategy	Trainable Parameters
Exp 1	ResNet50	Frozen backbone	558,468
Exp 2	ResNet50	Last 50 layers fine-tuned	17,508,996
Exp 3	EfficientNetB0	Last 30 layers fine-tuned	1,858,020

Table F: Training configuration

Setting	Value
Framework	TensorFlow/Keras
Image size	224 × 224
Batch size	32
Optimizer	Adam
Loss function	Sparse categorical cross-entropy
Metrics	Accuracy, precision, recall, macro-F1
Callbacks	Early stopping and learning rate reduction

5.6 Evaluation Metrics

The main metrics were accuracy, precision, recall, F1-score, normalized confusion matrix, and McNemar's test. Macro-F1 was used because it gives equal importance to all classes. McNemar's test was used to compare paired predictions from two models and check whether the difference between classifiers was statistically significant.

6. Results and Analysis

6.1 Main Performance Comparison

Table 1: Main performance comparison of all experiments

Experiment	Model	Training Data	Accuracy	Macro-F1	Time
Exp 1	ResNet50 Frozen	Dataset 1	91.50%	91.30%	4.9 min
Exp 2	ResNet50 Fine-tuned	Dataset 1	93.88%	93.71%	10.2 min
Exp 3	EfficientNetB0 Proposed	Dataset 1 + Dataset 2	94.50%	94.34%	6.6 min

The frozen ResNet50 baseline achieved 91.50% accuracy. This shows that pretrained ImageNet features are useful for MRI classification, even when the backbone is not trained. The fine-tuned ResNet50 model improved accuracy to 93.88%, confirming that fine-tuning helped the model learn MRI-specific features. The proposed EfficientNetB0 model achieved the highest numerical accuracy, with 94.50% accuracy and 94.34% macro-F1 score.

Table 2: Improvement between experiments

Comparison	Accuracy Gain	Macro-F1 Gain
Exp 2 vs Exp 1	+2.38%	+2.41%
Exp 3 vs Exp 1	+3.00%	+3.04%
Exp 3 vs Exp 2	+0.62%	+0.63%

6.2 Class-wise Results

Table 3: Class-wise performance of frozen ResNet50

Class	Precision	Recall	F1-score	Support
glioma	96.12%	74.25%	83.78%	400
meningioma	81.58%	93.00%	86.92%	400
notumor	94.12%	100.00%	96.97%	400
pituitary	96.34%	98.75%	97.53%	400
Macro average	92.04%	91.50%	91.30%	1600

Table 4: Class-wise performance of fine-tuned ResNet50

Class	Precision	Recall	F1-score	Support
glioma	98.15%	79.50%	87.85%	400
meningioma	89.12%	96.25%	92.55%	400
notumor	91.53%	100.00%	95.58%	400
pituitary	98.03%	99.75%	98.88%	400
Macro average	94.21%	93.88%	93.71%	1600

Table 5: Class-wise performance of proposed EfficientNetB0 on Dataset 1 test set

Class	Precision	Recall	F1-score	Support
glioma	99.07%	79.50%	88.21%	400
meningioma	87.95%	98.50%	92.92%	400
notumor	93.68%	100.00%	96.74%	400
pituitary	99.01%	100.00%	99.50%	400
Macro average	94.92%	94.50%	94.34%	1600

The proposed model achieved the best overall numerical result. Pituitary and notumor were classified very strongly. Glioma still had the lowest F1-score, mainly because its recall was lower than the other classes.

6.3 Efficiency Analysis

Table 6: Efficiency comparison

Model	Trainable Parameters	Training Time	Accuracy
ResNet50 Fine-tuned	17,508,996	10.2 min	93.88%
EfficientNetB0 Proposed	1,858,020	6.6 min	94.50%

EfficientNetB0 used far fewer trainable parameters than fine-tuned ResNet50. It also trained faster and achieved slightly higher accuracy. This is one of the strongest findings of the project because it shows a better accuracy-efficiency trade-off.

6.4 Held-Out Dataset 2 Evaluation

Table 7: Proposed EfficientNetB0 result on Dataset 2 test split

Model	Test Set	Accuracy	Macro-F1	Evaluated Images
EfficientNetB0 Proposed	Dataset 2 held-out test split	98.67%	98.58%	1280

The Dataset 2 test folder contained 1,311 images. The evaluation report used 1,280 images because the fixed-batch TensorFlow pipeline used batch size 32. This kept batch shapes stable during evaluation.

Table 8: Class-wise performance on Dataset 2 test split

Class	Precision	Recall	F1-score	Support
Glioma	99.65%	95.33%	97.44%	300
Meningioma	95.89%	99.02%	97.43%	306
No-tumor	100.00%	100.00%	100.00%	405
Pituitary	98.90%	100.00%	99.45%	269
Macro average	98.61%	98.59%	98.58%	1280

This result shows strong performance on the additional dataset test split. However, Dataset 2 training images were included in the proposed method's training process, so this is not a strict cross-domain test. A stricter cross-dataset test would train only on Dataset 1 and test on Dataset 2 without using Dataset 2 during training.

6.5 Visual Results

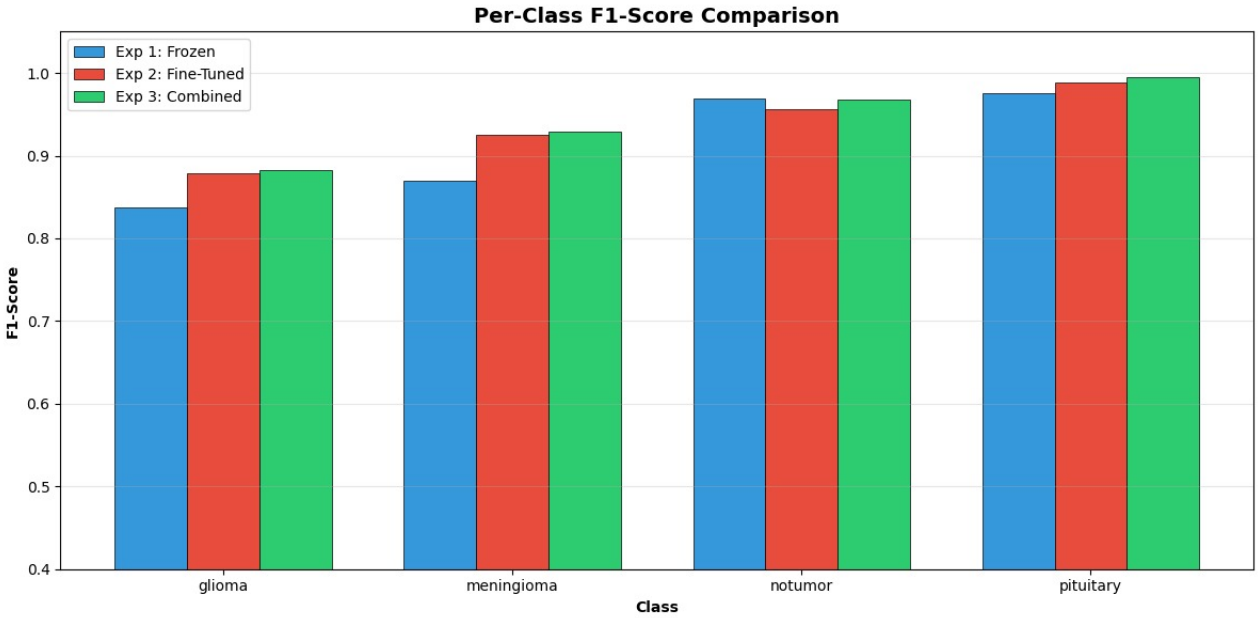

Figure 1: Per-class F1-score comparison across frozen ResNet50, fine-tuned ResNet50, and proposed EfficientNetB0.

The per-class F1 comparison shows that fine-tuning improved performance over the frozen baseline, especially for glioma and meningioma. The proposed EfficientNetB0 model achieved the best F1-score for glioma, meningioma, and pituitary. The notumor class remained strong across all experiments.

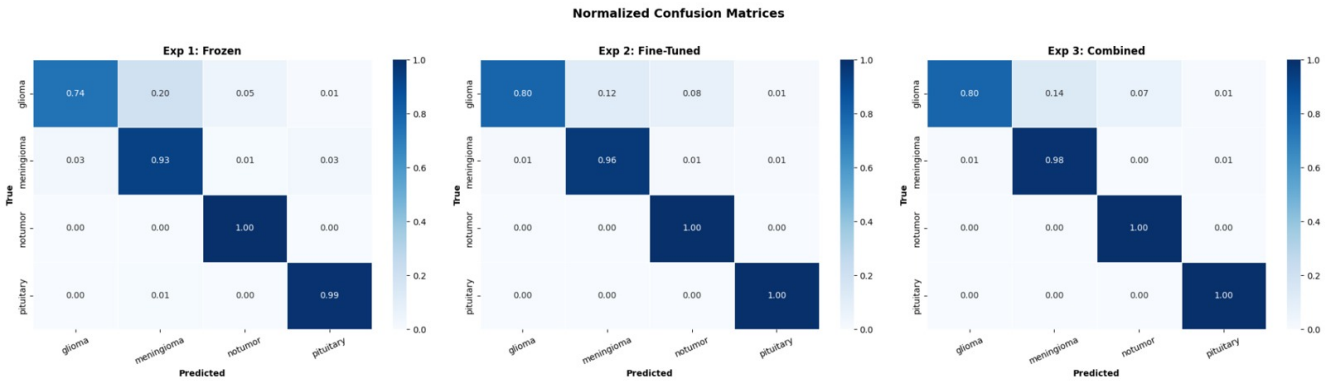

Figure 2: Normalized confusion matrices for frozen ResNet50, fine-tuned ResNet50, and proposed EfficientNetB0.

The confusion matrices show that most predictions were correct. The strongest performance was observed for notumor and pituitary classes. The main remaining confusion was between glioma and meningioma.

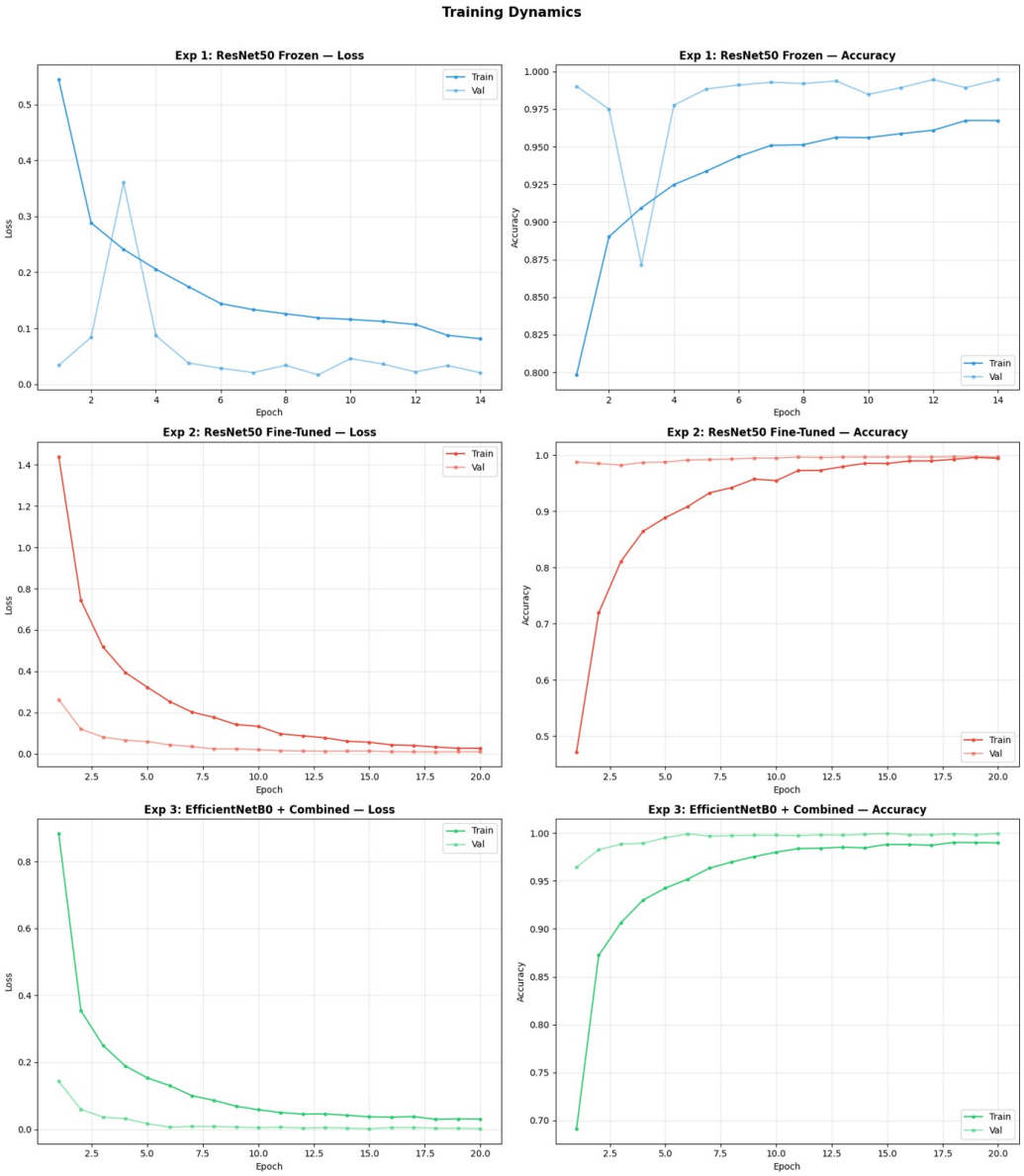

Figure 3: Training and validation loss/accuracy curves for all three experiments.

The training curves show that the models learned useful patterns during training. The proposed EfficientNetB0 model trained efficiently and reached strong validation performance in less time than fine-tuned ResNet50. The validation accuracy was very high, so the separate test set is treated as the main result.

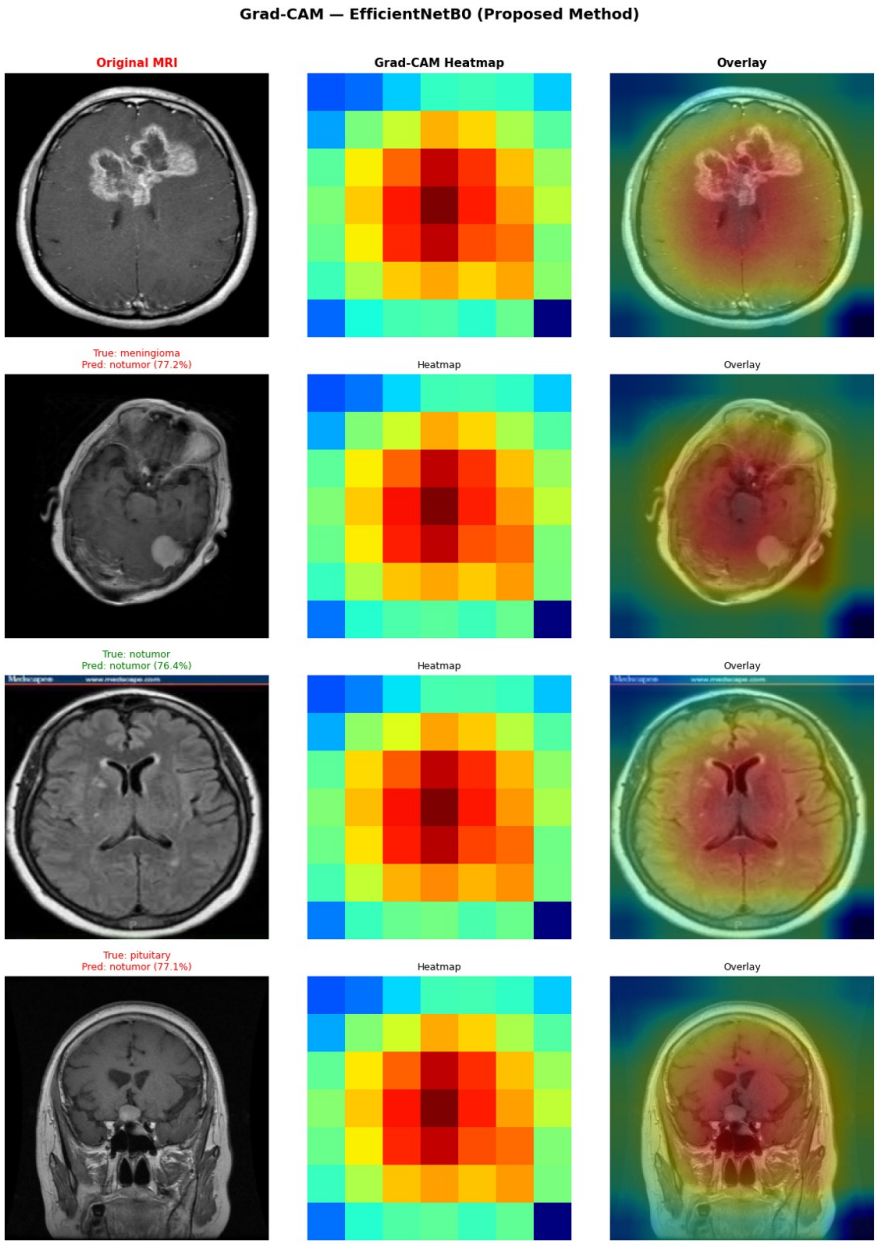

Figure 4: Grad-CAM visualizations for the proposed EfficientNetB0 model, including correct and difficult prediction examples.

The Grad-CAM visualizations help inspect model behavior. In some cases, the highlighted regions are near important brain regions. In other cases, the heatmaps are broad or the prediction is incorrect. Therefore, Grad-CAM should be treated as a qualitative explanation tool and not as medical proof.

6.6 Statistical Significance Testing

Table 9: McNemar's test results

Comparison	Chi-square	p-value	Significant at 0.05?
Exp 1 vs Exp 2	21.3906	0.000004	Yes
Exp 1 vs Exp 3	36.8167	0.000000	Yes
Exp 2 vs Exp 3	1.9286	0.164915	No

The test shows that fine-tuned ResNet50 significantly improved over frozen ResNet50. The proposed EfficientNetB0 model also significantly improved over frozen ResNet50. However, the improvement of EfficientNetB0 over fine-tuned ResNet50 was not statistically significant because the p-value was greater than 0.05. Therefore, the proposed model achieved the best numerical result, but it cannot be claimed as statistically better than the fine-tuned ResNet50 baseline.

A paired t-test was also performed on per-class F1 scores, but this result should be interpreted carefully because there were only four classes. McNemar's test is more appropriate here because it compares individual paired predictions.

7. Discussion

The experiments show that transfer learning is useful for brain tumor MRI classification. Even the frozen ResNet50 baseline achieved strong performance. This means that pretrained CNN features can help in MRI classification, even though MRI images are different from natural ImageNet images.

Fine-tuning improved the ResNet50 result. This makes sense because the model was allowed to update deeper layers and learn features more specific to MRI images. The accuracy improved from 91.50% to 93.88%, and the macro-F1 score improved from 91.30% to 93.71%.

The proposed EfficientNetB0 model achieved the highest numerical performance. One possible reason is that EfficientNetB0 is an efficient architecture that can learn strong image features with fewer trainable parameters. Another possible reason is the use of an additional dataset. More training data can expose the model to more variation in MRI images.

The efficiency result is important. Fine-tuned ResNet50 used 17,508,996 trainable parameters, while EfficientNetB0 used only 1,858,020 trainable parameters. EfficientNetB0 also trained faster. This means the proposed model gave a better balance between accuracy and resource usage.

However, the improvement over fine-tuned ResNet50 was small. The McNemar test showed that this improvement was not statistically significant. Because of this, the result should not be overclaimed. It is better to say that EfficientNetB0 is a promising and efficient alternative rather than saying it clearly outperformed ResNet50.

The class-wise results show that glioma remained the most difficult class. The proposed model achieved high precision for glioma but lower recall. This means that when the model predicted glioma, it was usually correct, but it still missed many actual glioma cases. Meningioma was sometimes confused with glioma, which suggests that these two classes share similar visual patterns.

The Dataset 2 held-out result was very strong, but it should also be explained carefully. Since Dataset 2 training images were included in training, the Dataset 2 test result does not fully prove cross-domain generalization. A stronger experiment would train on Dataset 1 only and test on Dataset 2, or train on Dataset 2 only and test on Dataset 1.

Grad-CAM added an explainability part to the work. It helped visualize which parts of the MRI image influenced the prediction. This is useful in medical AI because predictions should not be treated like a black box. Still, Grad-CAM is only a support tool. Medical expert validation would be needed before using such a model in a real clinical setting.

The main limitation of this experiment is that the proposed method changed two things at the same time: the model architecture and the training data. Because of this, it is hard to know exactly whether the improvement came from EfficientNetB0, from additional data, or from both. Future work should include an ablation experiment where EfficientNetB0 is trained only on Dataset 1.

Another limitation is that the datasets are curated 2D MRI slices, not full 3D patient-level MRI studies. In real clinical use, patient-level validation would be more important. The experiment also used a single random seed. K-fold cross-validation would provide a more robust performance estimate.

Overall, the proposed method is useful because it adds three improvements to the reproduced baseline: a more efficient model backbone, additional dataset training, and explainability using Grad-CAM. The results are not perfect, but they provide a meaningful experimental expansion beyond the reproduced baseline.

8. Conclusion

This assignment extended a reproduced ResNet50-based brain tumor MRI classification pipeline through new experiments. Three models were compared: frozen ResNet50, fine-tuned ResNet50, and EfficientNetB0 trained with combined data from two MRI datasets.

The results showed that fine-tuning improved performance compared to the frozen baseline. The proposed EfficientNetB0 model achieved the highest numerical result, with 94.50% accuracy and 94.34% macro-F1 score on the Dataset 1 test set. It also achieved strong performance on the held-out Dataset 2 test split.

A major strength of the proposed method was efficiency. EfficientNetB0 used fewer trainable parameters and required less training time than fine-tuned ResNet50, while still achieving slightly better numerical accuracy.

However, the improvement over fine-tuned ResNet50 was not statistically significant according to McNemar's test. Therefore, the proposed method should be described as an efficient and promising extension, not as a fully proven superior model.

The main remaining weakness was confusion between glioma and meningioma. Future work should include EfficientNetB0 trained only on Dataset 1, true cross-dataset testing, k-fold cross-validation, class-balanced loss functions, and medical expert validation of Grad-CAM heatmaps.

9. References

[1] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2016.

[2] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," Proceedings of the IEEE International Conference on Computer Vision, 2017.

[3] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," International Conference on Machine Learning, 2019.

[4] I. Goodfellow, Y. Bengio, and A. Courville, Deep Learning. MIT Press, 2016.

[5] M. Nickparvar, "Brain Tumor MRI Dataset," Kaggle Dataset.

[6] M. Abouali, "MRI Brain Tumor Dataset 4-Class / BT-MRI Dataset," Kaggle Dataset.

[7] TensorFlow Keras Applications Documentation, "ResNet50 and EfficientNetB0 Pretrained Models."

[8] M. M. Zahoor et al., "Brain Tumor MRI Classification Using Deep Residual CNN," 2024.

[9] F. J. Dorfner et al., "Deep Learning for Brain Tumor Analysis in MRI," 2025.

[10] Add the exact original reproduced paper reference here if required by the instructor.

Code Repository and Experiment Documentation

The updated code repository should include the final notebook, experiment results CSV, saved figures, README file, dataset links, setup instructions, and model training/evaluation code.

Repository Link: Add GitHub repository link here.

The README should briefly explain how to run the notebook, what datasets are needed, and where the output figures are saved.